

Indian Institute of Technology Gandhinagar
Course Plan
EE 323 – Digital Signal Processing (Semester 1, 2026-27)

Week	Week Starting On	Proposed Activity
1	03 Aug 2026	Signals and Systems - Review
2	10 Aug 2026	Sampling, Z-transform, Pole-Zero Placement
3	17 Aug 2026	Discrete Fourier Transform, Discrete Cosine Transform, Time Frequency Analysis
4	24 Aug 2026	Fast Fourier Transform, The Goertzel Algorithm, Chirp-z Transform
5	31 Aug 2026	Overlap Add/Overlap Save Method
6	07 Sep 2026	Filter Structures, IIR Filter Design
7	14 Sep 2026	FIR Filter Design, Polyphase Filters
8	21 Sep 2026	Examination I
9	28 Sep 2026	Fourier Slice Theorem, Radon Transform
10	05 Oct 2026	Multi-rate Signal Processing
11	12 Oct 2026	Adaptive Signal Processing, Array Signal Processing & Beamforming
12	19 Oct 2026	Mid-Semester Break
13	26 Oct 2026	Linear Prediction, Levinson-Durbin Algorithm
14	02 Nov 2026	Power Spectrum Estimation
15	09 Nov 2026	Finite Word Length Effects, Algorithms to Architecture
16	16 Nov 2026	Cepstrum Analysis and Speech Processing, DSP Algorithms for Multimedia, Revision

Expected Learning Outcomes

On successful completion of this course, we will have a better understanding of signals we deal with in our daily life. In addition, we will be able to design and implement systems that can process these signals.

Text Books

1. Manolakis, D.G. and Ingle, V.K., *Applied Digital Signal Processing: Theory and Practice*. Cambridge University Press, 2011.
2. Tan, L. and Jiang, J., *Digital Signal Processing: Fundamentals and Applications*. Academic press, 2018.
3. Lyons, R.G., *Understanding Digital Signal Processing*, 3/E. Pearson Education India, 2010.

References

1. Oppenheim, A.V., MIT OCW, *Digital Signal Processing*, 1987
2. Diniz, P.S.R, *Adaptive Filtering*, Springer, 2020

Mark Distribution

- Assignments - 20
- Class Participation - Upto 10
- Examination I - 25
- Examination II - 25
- Quiz I - 10
- Quiz II - 10